Practice Questions

1.) Evaluate the integral $\int_0^1 \int_1^2 x(x+y)\,dy\,dx$

Soln

$$= \int_0^1 \int_1^2 x(x+y)\,dy\,dx$$

$$= \int_0^1 \int_1^2 (x^2+xy)\,dy\,dx$$

$$= \int_0^1 \left[x^2y + \frac{xy^2}{2}\right]_1^2 dx$$

$$= \int_0^1 \left[\left(2x^2+\frac{4x}{2}\right)-\left(x^2+\frac{x}{2}\right)\right] dx$$

$$= \int_0^1 \left[2x^2+2x-x^2-\frac{x}{2}\right] dx$$

$$= \int_0^1 \left(x^2+\frac{3x}{2}\right) dx$$

$$= \int_0^1 \left(x^2+\frac{3x}{2}\right) dx$$

$$= \left[\frac{x^3}{3}+\frac{3x^2}{2\times 2}\right]_0^1$$

$$= \left[\frac{x^3}{3}+\frac{3x^2}{4}\right]_0^1$$

$$= \left[\left(\frac{1}{3}+\frac{3}{4}\right)-(0+0)\right]$$

$$= \frac{13}{12}$$ //

2.) Using double integration find the area enclosed by the curves $y = 2x^2$ and $y^2 = 4x$.

Soln

To find intersection points,

$$y = 2x^2$$

$$y^2 = 4x$$

$$\boxed{x = y^2/4}$$

$$y = 2\left(\frac{y^2}{4}\right)^2$$

$$y = \cancel{2}\left(\frac{y^4}{\cancel{16}8}\right)$$

$$8y = y^4$$

$$8y - y^4 = 0$$

$$y^4 - 8y = 0$$

$$y(y^3 - 8) = 0$$

$$\boxed{y = 0} \quad \boxed{y = 2}$$

When, $\boxed{y = 0}$

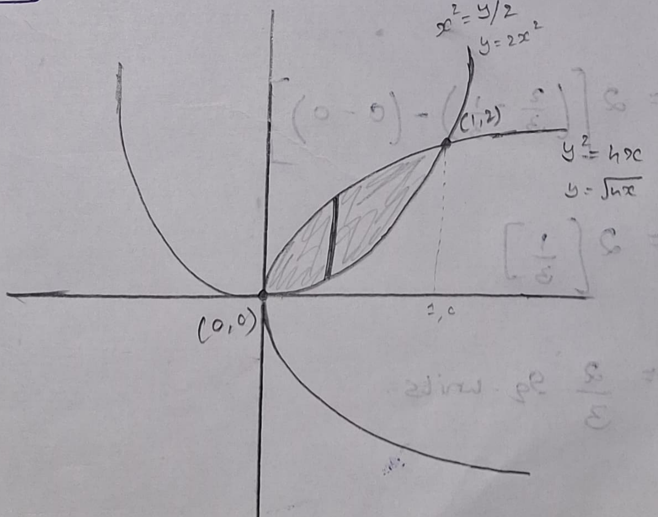
$\boxed{x = 0}$

When, $\boxed{y = 2}$

$x = y^2/4$

$x = 4/4$

$\boxed{x = 1}$



$$\boxed{\text{Area} = \iint dy\,dx}$$

$$\text{Area}(A) = \int_0^1 \int_{2x^2}^{\sqrt{4x}} dy\,dx$$

$$A = \int_0^1 [y]_{2x^2}^{\sqrt{4x}}\,dx$$

$$= \int_0^1 (\sqrt{4x} - 2x^2)\,dx$$

$$= \int_0^1 (2\sqrt{x} - 2x^2)\,dx$$

$$= 2\int_0^1 (\sqrt{x} - x^2)\,dx$$

$$= 2\int_0^1 (x^{1/2} - x^2)\,dx$$

$$= 2\left[\frac{x^{3/2}}{3/2} - \frac{x^3}{3}\right]_0^1$$

$$= 2\left[\frac{2x^{3/2}}{3} - \frac{x^3}{3}\right]_0^1$$

$$= 2\left[\left(\frac{2}{3} - \frac{1}{3}\right) - (0 - 0)\right]$$

$$= 2\left[\frac{1}{3}\right]$$

$$= \frac{2}{3} \text{ sq. units.}$$

3.) Evaluate the integral $\iint_R xy \, dx\,dy$ where R is the domain bounded by the x-axis, ordinate $x = 2a$ and the curve $x^2 = 4ay$.

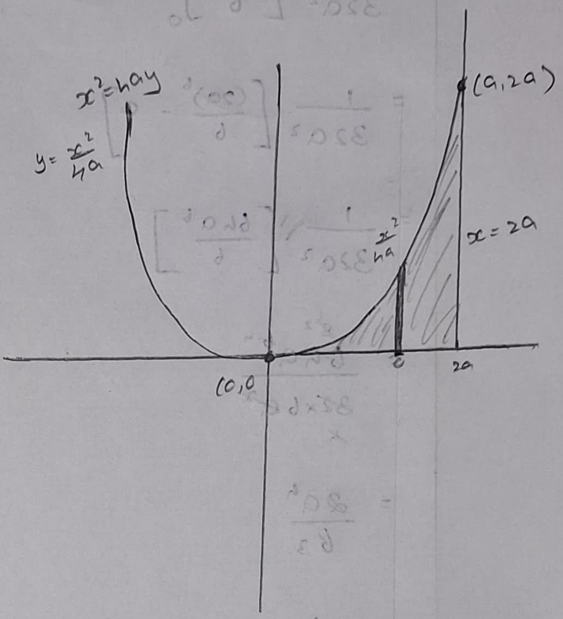
Soln Intersection Point,

$$x^2 = 4ay$$

$$(2a)^2 = 4ay$$

$$4a^2 = 4ay$$

$$\boxed{y = a}$$



when, $\boxed{y = a}$

$$x^2 = 4a(a)$$

$$x^2 = 4a^2$$

$$x = \sqrt{4a^2}$$

$$\boxed{x = 2a}$$

$$= \int_0^{2a} \int_0^{\frac{x^2}{4a}} xy \, dy \, dx$$

$$= \int_0^{2a} \left[\frac{xy^2}{2} \right]_0^{x^2/4a} dx$$

$$= \int_0^{2a} \left[\frac{x\left(\frac{x^2}{4a}\right)^2}{2} - 0 \right] dx$$

$$= \int_0^{2a} \left[\frac{x \frac{x^4}{16a^2}}{2} \right] dx$$

$$= \int_0^{2a} \left(\frac{x^5}{32a^2} \right) dx$$

$$= \frac{1}{32a^2}$$

$$= \frac{1}{32a^2} \int_0^{2a} x^5 \, dx$$

$$= \frac{1}{32a^2} \left[\frac{x^6}{6} \right]_0^{2a}$$

$$= \frac{1}{32a^2} \left[\frac{(2a)^6}{6} - 0 \right]$$

$$= \frac{1}{32a^2} \left[\frac{64a^6}{6} \right]$$

$$= \frac{\cancel{64}^{2} a^{\cancel{6}^{4}}}{\cancel{32}_{1} \times 6 \cancel{a^2}}$$

$$= \frac{2a^4}{\cancel{6}_{3}}$$

$$= \frac{a^4}{3}$$